Assignment 6

Q1 (a)

$$
\begin{aligned}
-1(1+e^{-x})^{-2} &= \frac{e^{-x}}{1+e^{-x}}\frac{1}{1+e^{-x}} \\
&= \frac{e^{-x}}{1+e^{-x}}\frac{1}{1+e^{-x}}
\end{aligned} \tag{1}
$$

but notice

$$\frac{e^{-x}}{1+e^{-x}} = 1 - \frac{1}{1+e^{-x}} \tag{2}$$

$$\frac{d}{dx}\sigma(x) = -1(1+e^{-x})^{-2} = \frac{1}{1-e^{-x}}\left(1 - \frac{1}{1+e^{-x}}\right) \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial a^{[2]}} = -\left(\frac{y}{a^{[2]}} - (1-y)\frac{1}{1-a^{[2]}}\right) \tag{4}$$

(b)

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial z^{[2]}} &= \frac{\partial L}{\partial a^{[2]}}\frac{\partial a^{[2]}}{\partial z^{[2]}} \\
&= -\left(\frac{y}{a^{[2]}} - (1-y)\frac{1}{1-a^{[2]}}\right)\left(\sigma(z^{[2]})(1-\sigma(z^{[2]}))\right) \\
&= -\left(\frac{y}{\sigma(z^{[2]})} - (1-y)\frac{1}{1-\sigma(z^{[2]})}\right)\left(\sigma(z^{[2]})(1-\sigma(z^{[2]}))\right) \\
&= \sigma(z^{[2]}) - y \\
&= a^{[2]} - y
\end{aligned} \tag{5}
$$

(c)

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial W^{[2]}} &= \partial_{z^{[2]}}\mathcal{L}\cdot\partial_{W^{[2]}}z^{[2]} \\
&= \left(a^{[2]} - y\right)\left(a^{[1]}\right)^T
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
\frac{\partial \mathcal{L}}{\partial b^{[2]}} &= \partial_{z^{[2]}}\mathcal{L}\cdot\partial_{b^{[2]}}z^{[2]} \\
&= \left(a^{[2]} - y\right)
\end{aligned} \tag{7}
$$

note

$$\frac{\partial \mathcal{L}}{\partial z^{[2]}} \in \mathbb{R}$$

$$\frac{\partial \mathcal{L}}{\partial W^{[2]}} \in \mathbb{R}^{1\times h}$$

$$\frac{\partial \mathcal{L}}{\partial b^{[2]}} \in \mathbb{R}^{1\times 1}$$

(d)

$$\frac{\partial \mathcal{L}}{\partial a^{[1]}} = \left(\frac{\partial z^{[2]}}{\partial a^{[1]}}\right)^T\frac{\partial \mathcal{L}}{\partial z^{[2]}} = (W^{[2]})^T\left(a^{[2]} - y\right) \tag{8}$$

$$\frac{\partial \mathcal{L}}{\partial z^{[1]}} = \left[(W^{[2]})^T \left(a^{[2]} - y\right)\right] \odot \mathrm{ReLU}'(z^{[1]}) \tag{9}$$

let $\odot$ be the element wise product (this lets $\partial_{z^{[1]}}\mathcal{L}$ be 0 when $z^{[1]}$ is less than zero where needed) $\odot$ is also known as the hadamar product.

(e)

$$\frac{\partial \mathcal{L}}{\partial W^{[1]}} = \frac{\partial \mathcal{L}}{\partial z^{[1]}} x^T = \left(\left[(W^{[2]})^T \left(a^{[2]} - y\right)\right] \odot \mathrm{ReLU}'(z^{[1]})\right) x^T \tag{10}$$

$$\frac{\partial \mathcal{L}}{\partial b^{[1]}} = \frac{\partial \mathcal{L}}{\partial z^{[1]}} \cdot 1 \tag{11}$$

Q2 (a)

$$f(u) = u(1-u) = u - u^2 \tag{12}$$

but note

$$\frac{df}{du} = 1 - 2u = 0 \implies u = \frac{1}{2} \tag{13}$$

and note that sigmoid is increasing so this is a maximum.

$$\sigma(x) = \frac{1}{2} \implies x = 0 \tag{14}$$

and

$$\sigma'(0) = \frac{1}{4} \tag{15}$$

1. note that

$$\frac{\partial \mathcal{L}}{\partial z^{[\ell]}} = \frac{\partial \mathcal{L}}{\partial z^{[L]}} \prod_{k=\ell+1}^{L} \frac{\partial z^{[k]}}{\partial z^{[k-1]}} \tag{16}$$

Since $z^{[k]} = W^{[k]} a^{[k-1]} + b^{[k]}$ and $a^{[k-1]} = \sigma(z^{[k-1]})$, we have $\frac{\partial z^{[k]}}{\partial z^{[k-1]}} = W^{[k]}\sigma'(z^{[k-1]})$. Substituting this back gives:

$$\frac{\partial \mathcal{L}}{\partial z^{[\ell]}} = \frac{\partial \mathcal{L}}{\partial z^{[L]}} \prod_{k=\ell+1}^{L} W^{[k]}\sigma'(z^{[k-1]}) \tag{17}$$

but each term in the product is bounded by $\frac{1}{4}$. taking the norm on both sides gives

$$\frac{\partial \mathcal{L}}{\partial z^{[\ell]}} \leq \left(\frac{1}{4}\right)^{L-\ell} \cdot \frac{\partial \mathcal{L}}{\partial z^{[L]}} \cdot \prod_{k=\ell+1}^{L} \parallel W^{[k]} \parallel \tag{18}$$

computing for $L = 10, \ell = 1$,

$$\left(\frac{1}{4}\right)^9 = \frac{1}{262,144} \approx \mathbf{3.81 \times 10^{-6}}$$

2.

$$\mathrm{ReLU}'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

this is less of a problem since each relu is slope 1 and doesnt suffer from the same "vanishing gradient" problem. the norm does not decrease every layer like sigmoid does.

3. if z is constantly less than zero, then relu function outputs 0 during backprop.

   one can remedy this by using many variants of relu like leaky relu.